

Molecular sex identification of red panda (*Ailurus fulgens*) suitable for noninvasive genetic studies

Arun Kumar¹ · Bhupen Roka² · Upashna Rai² · P. Anuradha Reddy¹

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Abstract Molecular sexing of wild animals is feasible with differential amplification of X- and Y-homologous regions of various genes. We present a simple and robust technique for sex identification of red panda suitable for noninvasive studies based on sex-specific fragment size polymorphism in amelogenin (*AMEL*) gene. Sequence analysis of *AMELX* and *AMELY* revealed an 18-bp deletion in Y fragment which was used to differentially amplify 217 bp of X and 199 bp of Y fragments. The designed primer pair is very sensitive and exhibits polymerase chain reaction (PCR) success rates of 89 and 78 % at 0.1 and 0.01 ng template DNA concentrations, respectively. Further, we amplified poor quality faecal DNA with 94 % success rate and average dropout rate of 15.6 %. In silico analysis shows that these primers can also be used to identify sex of a sympatric species, giant panda.

Keywords Red panda · Sex identification · Amelogenin · DNA

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✉ P. Anuradha Reddy
anuradha@ccmb.res.in

¹ CSIR-Centre for Cellular and Molecular Biology, Uppal Road, Hyderabad 500 007, India

² Padmaja Naidu Himalayan Zoological Park, Darjeeling 734101, India

Introduction

Noninvasive genetics has led to remarkable changes in population studies of endangered species (Beja-Pereira et al. 2009). This approach is especially useful in generating large-scale molecular data on elusive species without handling or even seeing them, from shed samples like hair, skin, feathers and faeces (Beja-Pereira et al. 2009; Reddy et al. 2012). Major challenges of working with such samples are low quality and quantity of DNA, genotyping errors and low amplification success rates (Taberlet et al. 1996). These issues are effectively addressed in studies which identify and quantify target DNA (Morin et al. 2001), use multi-tube method (Taberlet et al. 1996) and use multiplex DNA amplification (Arandjelovic et al. 2009), making noninvasive DNA analysis a robust method for assessing population parameters of wild animals.

Sex identification and sex ratio assessment provide crucial data on population dynamics of wild animals (Xu et al. 2010). Sex can be assessed by karyotyping, presence of Y-antigen, X-linked enzymes and polymerase chain reaction (PCR), of which PCR is a reliable, accurate and sensitive technique (Fernández Feijóo et al. 2010). PCR-based identification can be achieved by amplifying Y-specific SRY gene (Woods et al. 1999); however, an amplification failure can result in a false inference of a male as a female (Bidon et al. 2013). Internal mitochondrial controls can be used to address this problem (Kamimura et al. 1997), but there exists an inherent risk of preferential amplification, as mitochondrial sequences occur in higher copy numbers than Y-chromosome sequences (Durnin et al. 2006). Therefore, the most preferred technique is differential amplification of alleles located on homologous regions of sex chromosomes (Xu et al. 2010), such as zinc finger locus (*ZnF*) and amelogenin gene (*AMEL*) (Pilgrim et al. 2005). Length polymorphisms in *AMEL* and *ZnF* have been used in molecular sexing of free ranging animals like

Japanese black bear (*Ursus thibetanus japonicus*) (Yamamoto et al. 2002), felids (Pilgrim et al. 2005), giant panda (*Ailuropoda melanoleuca*) (Xu et al. 2007, 2010) and ursids (Pagès et al. 2008). Amplification-refractory mutation system (ARMS) is also used to differentially amplify X- and Y-chromosomal regions (Li et al. 2011).

Red panda (*Ailurus fulgens*) is a specialized carnivore that primarily feeds on bamboo and is distributed widely in the Himalaya and Hengduan mountains (Li et al. 2005; Hu et al. 2011; Wei et al. 2015). This endangered species is extremely difficult to monitor in the wild because of its elusive, wary and nocturnal behavior (Hodgson et al. 1847; Johnson et al. 1988; Holst 1989; Stevenson et al. 1989; Guo et al. 2011). Red panda is sympatric to giant panda, another important conservation flagship species, across large parts of its range. Despite being an important species, information on red panda from India is scanty mainly due to difficulty in studying this elusive and solitary animal, which dwells in mountainous ranges with dense forest cover. In addition to determining occupancy and population numbers of red panda, it is also crucial to determine sex ratio to understand its population dynamics, dispersal patterns and behavior (Durnin et al. 2006).

Here, we describe a simple and robust molecular sexing method based on sex-specific length polymorphism of amelogenin (*AMEL*) gene in red panda. We also test the efficiency of our primers for amplification at picogram concentrations of template DNA suitable for noninvasively collected samples.

Materials and methods

Sample collection and DNA extraction

Fresh faecal samples were collected from 15 red pandas housed at Padmaja Naidu Himalayan Zoological Park, Darjeeling, in sterile bottles with 90 % ethanol. Blood samples of seven red pandas were collected in EDTA-coated vacutainer tubes. All samples were transported to the laboratory within a month and stored at -20°C until extraction. DNA was isolated from blood by phenol-chloroform method (Sambrook et al. 1989) and from outer layer of faecal samples with QIAamp DNA stool mini kit (Qiagen).

Marker development and PCR analysis

Exon-5 of amelogenin gene of red panda was partially amplified as described by Yamamoto et al. (2002). *AMELX* fragment from females and *AMELX* and *AMELY* fragments from males were run on agarose gel, purified using QIAquick gel extraction kit (Qiagen) and sequenced by Big dye terminator cycle sequencing in ABI 3730 DNA Analyzer (Applied Biosystems). Sequences were analysed and edited with CodonCode Aligner v4.2 (GenBank accession numbers—

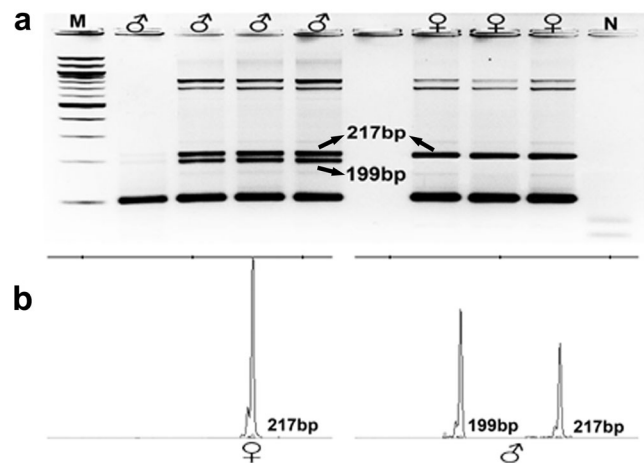


Fig. 1 Agarose gel (a) and automated (b) electropherograms of amelogenin locus in female (217 bp), and male (199 and 217 bp) red pandas. *M* 100-bp size marker, *N* negative control, ♀ female, ♂ male

KM017653 for *AMELX* and KM017654 for *AMELY*). These sequences were aligned and analysed for sequence polymorphism with MEGA v5.0 (Tamura et al. 2011) and ClustalW2 Online tool from EBI (<http://www.ebi.ac.uk/Tools/msa/clustalw2/>). Internal primer pair (RP-AmlF 5'-ACACTCCA TGACTCCAACCC-3' and RP-AmlR 5'-CTGTATGGGGAA CATCGGAG-3') was designed flanking the polymorphic region in *AMELX* and *AMELY* with Primer3 Plus (Untergasser et al. 2007) and was used to amplify 199 and 217 bp of *AMELY* and *AMELX*, respectively, with the following PCR conditions: initial denaturation at 95°C for 10 min, followed by 40 cycles of 95°C for 20 s, 54°C for 30 s, 72°C for 30 s, and final extension at 72°C for 10 min. PCR reaction mixture (15 μl) included 10 ng of template DNA, 1X Ex-*Taq* Buffer (Takara), 200 μM of each dNTP, 1.5 μg of BSA, 1.5 μM each of forward and reverse primer and 1 U of Ex-*Taq* polymerase (Takara). Since red panda is a specialized carnivore which feeds mainly on bamboo, contamination of faecal DNA with prey DNA is uncertain or possibly insignificant. However, as giant panda is often seen in sympatry with red panda, our primers were also tested for in silico PCR (Yu and Zhang 2011) of partial amelogenin sequences of giant panda (GenBank accession numbers EF635972, EF635973).

Table 1 Number of PCR replicates needed for different concentrations of blood DNA to obtain homozygous genotypes with 99.9 % certainty

Concentration (ng/ μl)	PCR success (%)	Drop rate (%)	Replicates needed
10	100	0	2
1	100	0	2
0.1	88.89	14.29	4
0.01	77.78	25	4
0.001	22.22	0	4

Blood genomic DNA of six red pandas (three males and three females) was serially diluted at 1:10 ratio from 10 to 0.001 ng/μl concentration and amplified with labeled RP-Aml primers in triplicates. In addition to agarose gel electrophoresis, PCR products were also electrophoresed in ABI 3730 DNA Analyzer (Applied Biosystems) with an internal LIZ size standard (Applied Biosystems). Alleles were scored with GeneMapper v3.2 software. Primers were finally tested for their ability to amplify DNA from faecal samples of 15 captive red pandas (ten males and five females).

Results and discussion

We describe a PCR method suitable for DNA-based sex identification of red panda from poor quality samples like faeces. Only one paper so far describes sex identification of red panda using ARMS PCR to detect sequence differences in X- and Y-homologous regions of zinc finger locus (*ZnF*) (Li et al. 2011). However, ARMS technique suffers an innate risk of preferential amplification of one amplicon over the other and is not suitable for molecular sexing (Xu et al. 2010). It is also our personal experience that ARMS technique requires pure genomic DNA like that obtained from blood and tissue, and is not suitable for noninvasive samples like faeces (present study, data not shown) and feathers (Reddy et al. 2007). It is simpler to use a single primer pair to detect length polymorphisms in X- and Y-homologous loci. Further, contrary to findings by Li et al. (2011), we obtained length polymorphism in homologous sequences of amelogenin (*AMEL*) gene in red panda (Fig. S1).

Approximately 430 bp of *AMELX* and 410 bp of *AMELY* were amplified and sequenced in red panda using the primers described by Yamamoto et al. (2002). Sequence alignment showed 18 and 63-bp deletions in *AMELY* of red panda and giant panda respectively (Fig. S1). Based on the length polymorphism between homologous amplicons, a new primer pair (RP-AmlF and RP-AmlR) was designed to amplify 217 and 199 bp of *AMELX* and *AMELY*, respectively, in red panda samples. Male samples produced amplicons of two sizes (199 and 217 bp), while female samples produced a single amplicon of 217 bp (Fig. 1a, b). The same primer pair, when tested in silico, amplified 214 and 151 bp of *AMELX* and *AMELY*, respectively, in giant panda (Fig. S1). This is advantageous as short PCR fragments (<250 bp) facilitate analysis of degraded DNA (Bidon et al. 2013). The diagnostic *AMELY* amplicon is smaller than *AMELX* amplicon, thereby avoiding misclassification of low quality male samples as females, as longer amplicons are more prone to amplification failure (Bidon et al. 2013). Size difference of 18 bp between *AMELX* and *AMELY* amplicons can be clearly resolved on 2.5 % agarose gel. However, in case of degraded DNA, amplicon sizes can also be determined by automated capillary electrophoresis.

Our primer pair is sensitive and could amplify template DNA concentrations of 0.1 ng with 89 % success and 0.01 ng with 78 % success when tested on blood samples (Table 1, Fig. S2). Dropout rate was higher for *AMELX* than the diagnostic *AMELY* amplicon, thereby confirming preferential amplification of the smaller Y fragment than the larger homologous X fragment. Dropout rate was also used to calculate the number of repetitions needed to obtain results with 99.99 % certainty for the homozygous (female) samples (Table 1). Sensitivity of these primers on low quality faecal DNA from 15 red pandas was good (Fig. S3), and we obtained 94.2 % PCR success with 15.6 % dropout rate in ten heterozygous (male) samples. Four PCR replications were needed to obtain results with 99.99 % certainty with faecal DNA samples.

Our newly developed PCR protocol was successfully applied to the captive red panda population in Padmaja Naidu Himalayan Zoological Park, and will facilitate sex identification of newborn animals in captivity. This technique can also be used to establish sex ratio of wild populations through analysis of noninvasive samples.

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Conflict of interest The authors declare that they have no competing interest.

Ethical approval Blood samples used in this study were acquired from the Padmaja Naidu Himalayan Zoological Park, Darjeeling, India, following the Padmaja Naidu Himalayan Zoological Park and Central Zoo Authority, Government of India, ethical guidelines and procedures. No animals were killed or captured as a result of this study.

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